THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If a pesticide kills all the frogs in a field, the number of insects will most likely: 1

(a) become constant **(b)** stay zero **(c)** increase **(d)** decrease

2. A student says the Earth's inner core is liquid because it is hotter than the outer core. This reasoning is wrong because: 1

(a) the inner core is gas **(b)** the outer core is cooler

(c) pressure raises the melting point **(d)** iron cannot be solid

3. A student sees in a concave mirror a diminished inverted real image. The object is: 1

(a) at C **(b)** at F **(c)** between F and P **(d)** beyond C

4. After 33 years the lunar and solar years differ by about: 1

(a) ten years **(b)** one month **(c)** one year **(d)** one day

5. Lichen is an association of: 1

(a) algae and animals **(b)** algae and fungi

(c) fungi and animals **(d)** bacteria and plants

6. A town draws more water from wells than rain recharges. In the long term: 1

(a) the water table falls **(b)** ozone increases

(c) the water table rises **(d)** wells overflow

7. Assertion (A): A convex lens can start a fire using sunlight. Reason (R): It converges parallel rays to a point. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): The Moon is visible at night. Reason (R): The Moon produces its own light. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. Assertion (A): Decomposers are essential for the recycling of nutrients. Reason (R): They convert dead organic matter into simple inorganic substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The Moon has large changes in surface temperature between day and night. Reason (R): The Moon has a thick atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. A clock's image in a plane mirror shows 10:10 in a mirror. The actual time is: 1

(a) 1:50 **(b)** 2:50 **(c)** 11:50 **(d)** 10:10

12. The new moon of 1 January 2024 is followed by a full moon after about: 1

(a) 15 days **(b)** 7 days **(c)** 60 days **(d)** 30 days

13. A lake in which the algae grow excessively after fertiliser run-off will show: 1

(a) a decrease in dissolved oxygen after the algae die
(b) increase in fishes
(c) clear water
(d) cool water

14. A planet has a thick CO2 atmosphere and a surface hot enough to melt lead. This planet is most like: 1

(a) Mars **(b)** Earth **(c)** Venus **(d)** the Moon

15. If the object distance from a convex mirror is increased, the image becomes: 1

(a) real **(b)** inverted
(c) larger **(d)** smaller and closer to F

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do herbivores form the largest biomass in animals but fewer in number than plants? 2

17. Why are mountain areas colder than the plains even though they are nearer the Sun? 2

18. A person walks away from a plane mirror at 3 m/s. Find the speed at which the image moves relative to the person. 2

19. If a 1.5 m stick has a 2 m shadow, find the height of a building with a shadow 40 m long at the same time. 2

20. Why can heavy use of pesticides in fields harm the birds that feed on insects there? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A farmer sprays an insecticide that kills all insects in a field. Predict three consequences. 3

22. Estimate the percentage of the Earth's radius that is made up of the crust given that the thickest continental crust is 70 km and Earth's radius is 6400 km. 3

23. Explain the use of a reflector behind a bulb in a torch with a diagram description. 3

24. India is 82.5 degrees E for IST. Find the time difference between a place at 75 degrees E and IST. 3

OR

State how the afforestation of degraded land restores the ecosystem. 3

25. Explain how the biosphere depends upon the other three spheres with examples. 3

OR

If a mirror is curved outwards, how do you predict the nature of the image and its size? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two plane mirrors are placed at right angles. A ray strikes the first mirror at an angle of incidence of 35 degrees. Find the angle of incidence at the second mirror and show that the emergent ray is parallel to the incident ray. 5

OR

Explain, with calculations, how the Gregorian calendar corrects for the 365.2422-day tropical year, listing the three rules for leap years, and test the years 1800, 1900, 2000, 2024 and 2100. 5

27. Analyse the food web: grass, grasshopper, rabbit, mouse, frog, snake, hawk. Identify all food chains, the trophic levels of the hawk, and the effect of removing the frogs. 5

OR

Analyse the data: Venus (460 degrees Celsius, thick CO2), Earth (15 degrees Celsius, thin atmosphere), Mars (-60 degrees Celsius, very thin). Explain the role of atmosphere and distance in these temperatures, and what Earth teaches about the greenhouse effect. 5

END OF QUESTION PAPER — Total: 50 Marks

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